

Question Formatting Cheat-Sheet

A quick reference for adding questions with **math**, **formatting**, and **images** in the LOCOMOTIVE admin panel.

Golden rule: never paste HTML. Everything below is safe, renders consistently on every device, and shows up correctly in tests. Use the **Preview** box under each field to check as you type.

1. The two math modes

All math goes between dollar signs. LaTeX inside is the same either way — only the size/placement changes.

Use	Wrap with	When to use
Inline math	<code>\$... \$</code>	Math <i>inside a sentence</i> or a short option, e.g. The value x^2 is...
Display math	<code>\$\$... \$\$</code>	A big standalone equation — large, centered, full-size fractions. Use this for the main expression in a question.

Example

- Type: Simplify $\frac{a^2}{b}$
- Type in an option: $\frac{7^5}{5^5}$

2. Fractions

Want	Type this
a fraction a/b	$\frac{a}{b}$
big centered fraction	$\frac{a}{b}$
fraction inside a fraction	$\frac{\frac{1}{2}}{3}$

The **a/b** button in the toolbar inserts the fraction template for you.

3. Powers & exponents

Want	Type this	Looks like
square	x^2	x^2
any power	x^{10}	x^{10} (always use { } for more than one character)
fractional power	$32^{3/5}$	$32^{(3/5)}$
power of a power	$(7^2)^{7/2}$	
scientific notation	6.02×10^{23}	6.02×10^{23}

Important: x^{10} shows as x^10 (only the first character rises). Always wrap in braces: x^{10} .

4. Subscripts (and chemistry)

Want	Type this	Looks like
subscript	<code>\$x_1\$</code>	x_1
multi-char subscript	<code>\$v_{\max}\$</code>	$v_{\max}$
water	<code>\$H_{20}\$</code>	H_2O
carbon dioxide	<code>\$CO_2\$</code>	CO_2
ion charge	<code>\$Na^{+}\$</code> , <code>\$SO_4^{2-}\$</code>	Na^+, SO_4^{2-}

5. Roots

Want	Type this	Looks like
square root	<code>\$\sqrt{x}\$</code>	$\sqrt{x}$
root of an expression	<code>\$\sqrt{x^2 + 1}\$</code>	$\sqrt{x^2 + 1}$
cube / nth root	<code>\$\sqrt[3]{x}\$</code>	$\sqrt[3]{x}$

The **√ button** in the toolbar inserts this.

6. Operators & symbols

Symbol	Type this		Symbol	Type this
× (times)	<code>\times</code>		±	<code>\pm</code>
· (dot)	<code>\cdot</code>		∓	<code>\mp</code>
÷	<code>\div</code>		∞	<code>\infty</code>
≤	<code>\leq</code>		≈	<code>\approx</code>
≥	<code>\geq</code>		≠	<code>\neq</code>
→	<code>\rightarrow</code>		° (degrees)	<code>^\circ</code>
Σ	<code>\sum</code>		∫	<code>\int</code>

(All of these go **inside** `$ ... $`.)

7. Greek letters

Inside `$ ... $` : `\alpha` α, `\beta` β, `\gamma` γ, `\theta` θ, `\pi` π, `\mu` μ, `\lambda` λ, `\Delta` Δ, `\Omega` Ω, `\sigma` σ.

Capitalize the command for the capital letter: `\delta` δ vs `\Delta` Δ.

8. Parentheses that grow with the content

For tall expressions, use `\left(` and `\right)` so the brackets stretch:

```
$$\left( \frac{a}{b} \right)^2$$
```

9. Plain words inside math

Use `\text{...}` so words aren't italicised like variables:

```
$v = \frac{\text{distance}}{\text{time}}$
```

The `</>` button inserts `\text{ }`.

10. Text formatting (outside math)

Want	Type this
bold	<code>**bold**</code>
<i>italic</i>	<code>*italic*</code>
superscript (non-math)	<code>^{\sup}</code> → e.g. <code>1^{\st}</code>
subscript (non-math)	<code>_{\sub}</code>

11. Images & diagrams

Click the **image button** in the toolbar to upload a diagram — it inserts the link for you. Or type it manually:

```
![description](https://url-to-image)
```

Use an image only when the content is genuinely a picture (a graph, a labelled diagram). For equations, always prefer math (`$....$`) — it's sharper, searchable, and scales on mobile.

12. Worked examples

A. A big expression as the question stem (from Math Test):

Simplify the given expression:

```
$$\frac{32^{\frac{3}{5}} \times 49^{\frac{7}{2}} \times 81^{\frac{6}{4}}}{25^{\frac{10}{4}} \times 64^{\frac{3}{6}} \times 9^3}$$
```

Its options:

Option A: $\frac{7^5}{5^5}$
Option B: $\frac{7^7}{5^7}$
Option C: $\frac{7^7}{5^5}$
Option D: $\frac{7^5 \times 2}{5^5}$
Option E: $\frac{7^7 \times 3^2}{5^5}$

B. Polynomial division answer:

Solve and choose the correct option:

$$\frac{x^3 - 5x^2 + 8x - 4}{x^2 - 3x + 2}$$

Option A: $x - 2 + \frac{4}{x^2 - 3x + 2}$

C. Chemistry:

Balance the reaction: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

13. Tips

- **Always watch the Preview box** under the field — if the math looks wrong, the LaTeX has a typo (usually a missing $\}$ or $\$$).
- Use $\$ \$ \dots \$$ for the **main equation**, $\$ \dots \$$ for math **inside options and sentences**.
- Every $\$$ needs a closing $\$$; every $\{$ needs a closing $\}$.
- Don't paste HTML or Word equations — retype using the syntax above so it renders everywhere.

Toolbar buttons: Σ = inline math, $\Sigma\Sigma$ = display math, a/b = fraction, x^n = exponent, $\sqrt{}$ = root, $\</>$ = plain text in math, **image** = upload a diagram.